

	NOTA TECNICA TECHNICAL REPORT		Núm. No.	ME-T-NT-120044
	Departamento Department	Methods	Avión Aircraft	Technologies
<u>Título/Title</u> Program RESULTANTE User's Manual Version 3				
<u>Palabras clave/Key words</u> MANUALS / LOADS / COMPUTER PROGRAMS			Clasificación acceso Access class P1 Registro de Revisiones Revision Record Pag. 2	
<u>Resumen/Summary</u> This report describes the use of the RESULTANTE program, version 3, and it includes examples to further clarify its use. This program is a Load Processor issued to add all loads applied on a Finite Elements Model created for MSC/NASTRAN. It calculates the resultant, in terms of force and moment, referred to a considered point, for a set of single load cases as well as for combined load cases.				
PROPIEDAD DE EADS CASA, Sociedad Unipersonal. Este documento no puede ser utilizado ni reproducido total o parcialmente sin la previa autorización escrita de la Dirección de Ingeniería y Tecnología de EADS-CASA. EADS CASA, Sociedad Unipersonal PROPERTY. This document shall neither be used nor completely or partially reproduced without previous written authorization by EADS-CASA Engineering and Technology Directorate.				
Distribución/Distribution	Realización/Execution		Fecha 1ª edición 1st issue Date May – 2012	
Archivo de Cálculo	Realizado Prepared		Firma Signature	
Jefe de la Subd. de Cálculo (*)	Darío Fernández Villalba			
Jefes de Depto. de Calculo (*)				
Jefes de Sección de Cálculo				
	Comprobado Checked	Gustavo Baños Lapuente		
(*) Cover sheet only	Aprobado Approved	Darío Fernández Villalba		

TABLE OF CONTENTS

1.- Technical sheet of the program.....	4
2.- Introduction.....	5
3.- Execution of the program.....	7
4.- Input data.....	8
4.1 General input data file.....	8
4.2 MSC/NASTRAN input files	12
5.- Output results.....	14
5.1 Output results file.....	14
5.2 Useful Grids, Forces and Moments files	16
5.3 Error and warning messages	17
6.- Windows application.....	21
7.- Examples	22
7.1 Example 1.....	22
7.2 Example 2.....	25
7.3 Example 3.....	28
7.4 Example 4.....	31

1.- Technical sheet of the program

Technical Note **ME-T-NT-120044**

<u>PROGRAM</u>	<u>REFERENCE</u>	<u>DATE</u>
RESULTANTE		2012

OBJECTIVE

This program is a Load Processor issued to add all loads applied on a Finite Elements Model, calculating the resultant referred to a considered point, for a set of single load cases as well as for combined load cases.

LANGUAGE

Fortran 77

COMPUTER

Sun Solaris

NUMBER OF LINES (APROX.)

2500

EXECUTION MODE

Interactive

AUTHOR

Darío Fernández Villalba

VERSION

3.0

ADDITIONAL DOCUMENTS

Due to the characteristics of this program, it is not necessary to issue a theoretical manual.

2.- Introduction

This report describes the use of the RESULTANTE program, version 3, and it includes examples to further clarify its use.

This program is a Load Processor issued to add all loads applied on a Finite Elements Model created for MSC/NASTRAN. It calculates the resultant, in terms of force and moment, referred to a considered point, for a set of single load cases as well as for combined load cases.

The required data are the following:

- Title of the problem.
- Origin point to calculate the resultant of forces and moments.
- Definition of the cut section, if any, to remove the loads applied on a part of the structure.
- Input MSC/NASTRAN data files:
 - File containing the set of considered GRID points.
 - File with the FORCE and MOMENT cards, for single load cases.
 - File with the LOAD cards, for combined load cases. This file is optional.

These input files have to be written in MSC/NASTRAN format, with the limitations explained in further chapters. Since the program reads in each moment the considered cards and discard the rest, all MSC/NASTRAN cards can be included in a single file.

- Factors to be applied on the resultant of forces and moments.
- Output results filename.

The results are provided in terms of resultant of Forces and Moments, referred to the considered point, for a set of single load cases as well as for combined load cases.

Differences between versions:

- Version 1.0:

Initial version developed in Fortran 77 for Digital VAX computer.

- Version 2.0:

The program is adapted to the new Sun Solaris computer.

- Version 2.1:

The maximum number of GRID points is increased from 5000 to 50000.

It is included the option to consider the whole structure and to project the resultant in swept exes.

- Version 2.1:

The functionality of the program is improved.

- Version 3.0:

Now the user can choose the language for dialogs on the screen and for the output results file, between Spanish or English.

The program source is restructured to allow the creation of a Dynamic Link Library DLL for Windows operating system.

The input data cannot be provided in interactive mode. They must be contained in a general data file, organized in a similar way as other programs.

In parallel with this, a Windows application has been developed as Graphic User Interface GUI for the program RESULTANTE. This application uses the same input files and provides the same output results. The input, output and calculation functions are performed by calling to the DLL, compiled from the same source as the official Fortran version.

3.- Execution of the program

Prior to the program execution, it is necessary to create the general input data file, as described in section 4. The program can be then executed by typing its **full path name**.

If the software symbol table has been previously assigned (consult with the System Manager in order to know how to assign this symbol table), it can be executed by typing the command:

```
$resultante
```

The program asks for the language to use and the name of the input and output files, as shown in the following example:

```
Idioma / Language:
=====

 1 ... ESPAÑOL
 2 ... ENGLISH

Idioma / Language: 2

Input filename: Example.dat

Resultante 3.0. (c) Airbus Military 2012
```

The CPU time for calculation depends on the number of MSC/NASTRAN cards to be processed and it is, in general, negligible.

It is possible to execute the program with a single command, including the input filename and the language as parameters. The above example can be executed as follows:

```
$resultante Example.dat 2
```

4.- Input data

This program requires a general input data file a set of input data files. These input files must contain the following types of MSC/NASTRAN cards:

- GRID points of the FEM where loads are applied.
- FORCE and MOMENT cards, defining the single load cases to be considered.
- LOAD cards (optional) defining the combined load cases.

4.1 General input data file

This file contains the names of the input and output files as well as all required parameters to perform the calculation of the resultant for the load cases applied on the FEM.

It basically contains data groups, numbered from 1 to 5, appearing in any order.

GROUP 1: Title of the problem.

Four lines containing the title of the problem to be solved. This group is organised as follows:

Line 1 -----> 1 First line of the title
 Line 2 -----> Second line of the title
 Line 3 -----> Third line of the title
 Line 4 -----> Fourth line of the title

GROUP 2: Origin point:

This group contains the coordinates of the origin point for calculation of the resultant, expressed in terms of a force and a moment, in each load case.

The structure of this group is the following:

Line 1 -----> 2 “Optional commentary”
 Line 2 -----> X_o Y_o Z_o

Where: (X_o,Y_o,Z_o) are the coordinates of the origin point.

GROUP 3: Cut section:

This group specifies the section that defines a cut in the model to avoid consider loads in a part of the structure.

The arrangement of this group depends on the type of the considered cut section, and it is the following:

Line 1 -----> 3 “Optional commentary”

Line 2 -----> Type

Line 3 -----> Parameters (One, three or none)

Where: Type = Type of cut section.

Parameters = Required parameters to define the cut section. They can be one, three or none. If no parameter is required, the third line has to be omitted.

The types of cut section as well as the required parameters are defined in the table below:

Type	Description	Parameters
0	There is no cut section. The whole structure will be considered.	None (*)
1	Plane $X = X_{min}$	X_{min}
2	Plane $Y = Y_{min}$	Y_{min}
3	Plane $Z = Z_{min}$	Z_{min}
4	Plane $Y - Y_a = m \cdot (X - X_a)$	X_a Y_a m
5	Plane $Z - Z_a = m \cdot (X - X_a)$	X_a Y_a m
6	There is no cut section. The whole structure will be considered, but loads will be projected on swept axes.	θ

(*) If Type=0, the third line of the group will be omitted.

The loads, forces and moments, considered by the program RESULTANTE to calculate the resultant, depend on the type of the cut section, as follows:

- Type = 0:

All loads (forces and moments) applied in **every GRID** point of the FEM will be considered to calculate the resultant.

- Type = 1:

The cut section is defined by the plane $X = X_{min}$. Only the loads applied in GRID points for which the X-coordinate is $X \geq X_{min}$, will be considered.

- Type = 2:

The cut section is defined by the plane $Y = Y_{min}$. Only the loads applied in GRID points for which the Y-coordinate is $Y \geq Y_{min}$, will be considered.

- Type = 3:

The cut section is defined by the plane $Z = Z_{min}$. Only the loads applied in GRID points for which the Z-coordinate is $Z \geq Z_{min}$, will be considered.

- Type = 4:

The cut section is defined by the plane $Y - Y_a = m \cdot (X - X_a)$. Only the loads applied in GRID points for which the Y-coordinate is $Y \geq Y_a + m \cdot (X - X_a)$, will be considered.

The resultant is then projected on swept axes:

$$\theta = \text{tg}^{-1}(m)$$

$$F_{x_\theta} = F_x \cdot \cos(\theta) + F_y \cdot \sin(\theta)$$

$$F_{y_\theta} = F_y \cdot \cos(\theta) - F_x \cdot \sin(\theta)$$

$$F_{z_\theta} = F_z$$

$$M_{x_\theta} = M_x \cdot \cos(\theta) + M_y \cdot \sin(\theta)$$

$$M_{y_\theta} = M_y \cdot \cos(\theta) - M_x \cdot \sin(\theta)$$

$$M_{z_\theta} = M_z$$

This option is ideal to be applied on lifting surfaces with swept angle for which the intention is cutting in sections coherent with the aerodynamic.

- Type = 5:

The cut section is defined by the plane $Z - Z_a = m \cdot (X - X_a)$. Only the loads applied in GRID points for which the Z-coordinate is $Z \geq Z_a + m \cdot (X - X_a)$, will be considered.

This option has been issued to analyse the loads applied on a fuselage, from an end, up to an oblique frame.

- Type = 6:

All loads (forces and moments) applied in **every GRID** point of the FEM will be considered to calculate the resultant.

The resultant is then projected on swept axes:

$$F_{x_\theta} = F_x \cdot \cos(\theta) + F_y \cdot \sin(\theta)$$

$$F_{y_\theta} = F_y \cdot \cos(\theta) - F_x \cdot \sin(\theta)$$

$$F_{z_\theta} = F_z$$

$$M_{x_\theta} = M_x \cdot \cos(\theta) + M_y \cdot \sin(\theta)$$

$$M_{y_\theta} = M_y \cdot \cos(\theta) - M_x \cdot \sin(\theta)$$

$$M_{z_\theta} = M_z$$

GROUP 4: MSC/NASTRAN input files:

This group contains the filenames of the input files containing the GRID, FORCE, MOMENT and LOAD cards in MSC/NASTRAN format.

Its arrangement is the following:

Line 1 -----> 4 “Optional commentary”
 Line 2 -----> File with GRID cards
 Line 3 -----> File with FORCE and MOMENT cards.
 Line 4 -----> File with LOAD cards.

If only single load cases are going to be considered, LOAD cards must not be specified. In such a case, the **fourth line must be a blank line**.

Two or the three of the above files can be the same file. For example, if the GRID cards and the LOAD cards are included in the same file, the second and the fourth lines will be identical.

GROUP 5: Output results:

The structure of this group is the following:

Line 1 -----> 5 “Optional commentary”
 Line 2 -----> Ff Fm
 Line 3 -----> Output results filename

Where: Ff and Fm are factors to be applied on the forces and moments of the resultant, respectively.

These factors are especially useful to perform a change of units. For example if input data are provided in “mm”, “N” and “N·mm” and the resultant is required in “kN” and “kN·m”, the factors must be:

$$Ff = 1.0E-3$$

$$Fm = 1.0E-6$$

4.2 MSC/NASTRAN input files

Three MSC/NASTRAN input files must be specified. They are:

- File containing the set of considered GRID points.
- File with the FORCE and MOMENT cards, for single load cases.
- File with the LOAD cards, for combined load cases. This file is optional.

These input files have to be written in MSC/NASTRAN format, with the limitations explained below. Since the program reads in each moment the considered cards and discard the rest, all MSC/NASTRAN cards can be included in a single file.

File of nodes (GRID cards):

This file contains all GRID cards to be accounted for. If a cut section has been specified (options 1 to 5), the program will select as “useful nodes” those ones that meet with the imposed condition and it will discard the rest.

This file can also contain another type of cards that the program will ignore, and it has to be issued with the following limitations:

- The free format of fields, separated by commas, is not supported. However, it is possible to use <Tab>.
- All nodes have to be defined in the global coordinate system. Every value different from blank or zero in the field #3 will cause the node not to be considered and, therefore, forces or moments applied in such nodes will be discarded. Warning messages in the output results file will be written notifying this fact.
- The GRID cards have to have single fields, 10 fields of 8 characters each one. GRID cards with fields in double precision (fields 2 to 5 of 16 characters) will be ignored.

File of the loads (FORCE and MOMENT cards):

This file contains all FORCE and MOMENT cards to be accounted for. If a cut section has been specified (options 1 to 5), the program will select as “useful loads” those ones applied on nodes that meet with the imposed condition and it will discard the rest.

This file can also contain another type of cards that the program will ignore, and it has to be issued with the following limitations:

- The free format of fields, separated by commas, is not supported. However, it is possible to use <Tab>.
- The components of forces and moments have to be defined in the global coordinate system. Every value different from blank or zero in the field #4 will cause the load not to be considered. Warning messages in the output results file will be written notifying this fact.

- If the three components of a FORCE (F_x , F_y , F_z) or of a MOMENT (M_x , M_y , M_z) are null, the program will write a warning in the output file.

File of combined load cases (LOAD cards).

This file defined the combined load cases and it can be omitted. If present, it contains all LOAD cards to be accounted for.

This file can also contain another type of cards that the program will ignore, and it has to be issued with the following limitations:

- The free format of fields, separated by commas, is not supported. However, it is possible to use <Tab>.
- The continuation cards must be just after the main LOAD card.

5.- Output results

The program RESULTANTE creates the following files:

- Output results file.
- Useful Grid file.
- Useful Forces and Moments file

5.1 Output results file

This is a text file containing the following information:

- Eco of the input data:
 - Title of the problem.
 - Origin point for calculation of the forces and moments of the resultant.
 - Definition of the cut section.
- Summary of cards in the MSC/NASTRAN input files:
 - For the file of GRID cards, total number of cards:
 - = Read
 - = Useful
 - = Erroneous
 - For the file of FORCE and MOMENT cards, total number of cards:
 - = Read
 - = Useful
 - = Erroneous
 - For the file of LOAD cards, if specified, total number of cards:
 - = Read
- Resultant force and moment for single load cases. All load cases are displayed sorted, in ascending order. For each one, it is shown:
 - Load case number.
 - Resultant force: F_x , F_y and F_z .
 - Resultant moment: M_x , M_y and M_z .
- Resultant force and moment for combined load cases. All load cases are displayed in the same order as in the LOADs file. For each one, it is shown:
 - Load case number.

- Denomination of the load case.
- Resultant force: F_x , F_y and F_z .
- Resultant moment: M_x , M_y and M_z .

If the file of combined load cases has not been provided, obviously this section will be omitted.

5.2 Useful Grids, Forces and Moments files

When a cut section has been defined (options 1 to 5 in the group 3 of the general input data file), two additional files are created by the program:

- Useful Grid file:

When this file is created, it contains all considered GRID cards that meet with the definition of the cut section.

The format of this file is the same as the input file with MSC/NASTRAN GRID cards. For this reason, it can be used as input for another run of the program.

The name of this file is obtained by concatenation of the output file, without tip, plus the string **“.UsefulGrids.txt”** (or **“.GridsUtiles.txt”** in Spanish).

- Useful Forces and Moments file:

When this file is created, it contains all considered FORCE and MOMENT cards, applied on nodes that meet with the definition of the cut section.

The format of this file is the same as the input file with MSC/NASTRAN FORCE and MOMENT cards. For this reason, it can be used as input for another run of the program.

The name of this file is obtained by concatenation of the output file, without tip, plus the string **“.UsefulForces.txt”** (or **“.ForcesUtiles.txt”** in Spanish).

5.3 Error and warning messages

When the program detects an error in the input data, it writes the corresponding message in the console and stops the execution. In other cases, a warning is provided in the output file and the program continues the execution. The error and warning messages that might appear are listed below:

Error messages. English output:

```

ERROR: The input file does not exist
ERROR: The input file cannot be opened
ERROR: Erroneous number of group data
ERROR: Group 1 (Title) missing in the input file
ERROR: Reading error or end of file when reading Group 1 (Title)
ERROR: Group 2 (Origin point) missing in the input file
ERROR: Reading error or end of file when reading Group 2 (Origin
point)
ERROR: Group 3 (Cut section) missing in the input file
ERROR: Reading error or end of file when reading Group 3 (Cut
section)
ERROR: Group 4 (NASTRAN files) missing in the input file
ERROR: Reading error or end of file when reading Group 4 (NASTRAN
files)
ERROR: Group 5 (Results) missing in the input file
ERROR: Reading error or end of file when reading Group 5 (Results)
ERROR: The input file with GRIDs does not exist
ERROR: The input file with GRIDs cannot be opened
ERROR: The file for useful GRIDs cannot be opened
ERROR: It is not possible to write in the file for useful GRIDs
ERROR: The input file with FORCES & MOMENTS does not exist
ERROR: The input file with FORCES & MOMENTS cannot be opened
ERROR: The file for useful FORCES & MOMENTS cannot be opened
ERROR: It is not possible to write in the file for useful FORCES &
MOMENTS
ERROR: The input file with LOADs does not exist
ERROR: The input file with LOADs cannot be opened
ERROR: Erroneous data in a LOAD card
ERROR: Erroneous continuation fiel in LOAD card
ERROR: The output results file cannot be opened
ERROR: It is not possible to write in the output results file
ERROR: Too many GRID points, exceeding the program capability
ERROR: Too many single load cases, exceeding the program capability
ERROR: Too many combined load cases, exceeding the program capability
ERROR: Too many load cases for a combined case, exceeding the program
capability
    
```

Error messages. Spanish output:

ERROR: No existe el fichero de entrada
 ERROR: No se puede abrir el fichero de entrada
 ERROR: Número de grupo de datos erróneo
 ERROR: Falta Grupo 1 (Título) en el fichero de entrada
 ERROR: Error de lectura o fin de fichero leyendo el Grupo 1 (Título)
 ERROR: Falta Grupo 2 (Punto de origen) en el fichero de entrada
 ERROR: Error de lectura o fin de fichero leyendo el Grupo 2 (Punto de origen)
 ERROR: Falta Grupo 3 (Sección de corte) en el fichero de entrada
 ERROR: Error de lectura o fin de fichero leyendo el Grupo 3 (Sección de corte)
 ERROR: Falta Grupo 4 (Ficheros NASTRAN) en el fichero de entrada
 ERROR: Error de lectura o fin de fichero leyendo el Grupo 4 (Ficheros NASTRAN)
 ERROR: Falta Grupo 5 (Resultados) en el fichero de entrada
 ERROR: Error de lectura o fin de fichero leyendo el Grupo 5 (Resultados)
 ERROR: No existe el fichero de entrada con GRIDS
 ERROR: No se puede abrir el fichero de entrada con GRIDS
 ERROR: No se puede abrir el fichero de GRIDS útiles
 ERROR: No se puede escribir en el fichero de GRIDS útiles
 ERROR: No existe el fichero de entrada con FORCES & MOMENTS
 ERROR: No se puede abrir el fichero de entrada con FORCES & MOMENTS
 ERROR: No se puede abrir el fichero de FORCES & MOMENTS útiles
 ERROR: No se puede escribir en el fichero de FORCES & MOMENTS útiles
 ERROR: No existe el fichero de entrada con LOADS
 ERROR: No se puede abrir el fichero de entrada con LOADS
 ERROR: Datos erróneos en una tarjeta LOAD
 ERROR: Campo de continuación erróneo en tarjeta LOAD
 ERROR: No se puede abrir el fichero de salida de resultados
 ERROR: No se puede escribir en el fichero de salida de resultados
 ERROR: Demasiados puntos GRID, excediendo la capacidad del programa
 ERROR: Demasiados casos de carga simple, excediendo la capacidad del programa
 ERROR: Demasiados casos de carga combinados, excediendo la capacidad del programa
 ERROR: Demasiados casos de carga para un caso combinado, excediendo la capacidad del programa

Warning messages. English output:

Coordinate system ***** not accepted. Ignored the line:

GRID nnnnnn ***** xxx.xx yyy.yy zzz.zz

Erroneous data in a GRID. Ignored the line:

GRID nnnnnn xxx.xx yyy.yy zzz.zz

Number of load case ***** erroneous. Ignored the line:

FORCE ***** nnnnnn Fx Fy Fz F

Auxiliary coordinate system ***** not accepted. Ignored the line:

FORCE cccc nnnnnn ***** Fx Fy Fz F

Erroneous data in a FORCE. Ignored the line:

FORCE cccc nnnnnn Fx Fy Fz F

Warning. In the following card, Fx, Fy and Fz are null:

FORCE cccc nnnnnn 0 0 0 F

In the following card, a force is applied on a nonexistent GRID point:

FORCE cccc nnnnnn Fx Fy Fz F

Number of load case ***** erroneous. Ignored the line:

MOMENT ***** nnnnnn Mx My Mz M

Auxiliary coordinate system ***** not accepted. Ignored the line:

MOMENT cccc nnnnnn ***** Mx My Mz M

Erroneous data in a MOMENT. Ignored the line:

MOMENT cccc nnnnnn Mx My Mz M

Warning. In the following card, Mx, My and Mz are null:

MOMENT cccc nnnnnn 0 0 0 M

In the following card, a moment is applied on a nonexistent GRID point:

MOMENT cccc nnnnnn Mx My Mz M

Warning messages. Spanish output:

Sistema de coordenadas ***** no aceptado. Ignorada la línea:

GRID nnnnnn ***** xxx.xx yyy.yy zzz.zz

Datos erróneos en un GRID. Ignorada la línea:

GRID nnnnnn xxx.xx yyy.yy zzz.zz

Número de caso de carga **** erróneo. Ignorada la línea:

FORCE **** nnnnnn Fx Fy Fz F

Sistema auxiliar de coordenadas ***** no aceptado. Ignorada la línea:

FORCE cccc nnnnnn ***** Fx Fy Fz F

Datos erróneos en una FORCE. Ignorada la línea:

FORCE cccc nnnnnn Fx Fy Fz F

Aviso: En la siguiente tarjeta, Fx, Fy y Fz son nulas:

FORCE cccc nnnnnn 0 0 0 F

En la siguiente tarjeta, se aplica una fuerza en un GRID inexistente:

FORCE cccc nnnnnn Fx Fy Fz F

Número de caso de carga **** erróneo. Ignorada la línea:

MOMENT **** nnnnnn Mx My Mz M

Sistema auxiliar de coordenadas ***** no aceptado. Ignorada la línea:

MOMENT cccc nnnnnn ***** Mx My Mz M

Datos erróneos en un MOMENT. Ignorada la línea:

MOMENT cccc nnnnnn Mx My Mz M

Aviso: En la siguiente tarjeta, Mx, My y Mz son nulas:

MOMENT cccc nnnnnn 0 0 0 M

En la siguiente tarjeta, se aplica un momento en un GRID inexistente:

MOMENT cccc nnnnnn Mx My Mz M

6.- Windows application

An application has been developed in Visual Basic for Windows environment. The original routines of the program for input data, calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore the same results will be obtained, with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed. For installation, please contact with the system manager.

The application can be executed by "double click" in the file **Resultante.exe**, located in the PC's server, or by means a **link** on the "desktop" or in the "start menu".

The input data must be typed in the application form or retrieved from an input file. The form, after loading the data corresponding to the example3, has the following appearance:

The screenshot shows the 'Resultante' application window with the title bar 'Resultante - C:\Temp\Example3.dat'. The window is divided into several sections:

- Title of the problem:** Example 3 for the program RESULTANTE.
- Cut section:** Defined by the plane: $Z - 3000 = 4.70463 * (X - 12540)$.
- Input data units:** [L]=mm [F]=N [M]=N.mm
- Output results units:** [F]=kN [M]=kN.m
- Input files of Nastran data and output file:**
 - GRIDs: Example_Grid.dat
 - FORCES: Example_Force.dat
 - LOADs: Example_Load.dat
 - Results: Example3.txt
- Origin for calculation of forces and moments:**
 - Xo(x,y,z): 12540.0, 0.0, 3000.0
- Cut section:** Defined by the plane: $Z - Za = m * (X - Xa)$
 - Xa: 12540.0
 - Za: 3000.0
 - m: 4.70463
- Factors on the resultant loads:**
 - Forces: 1.0000E-03
 - Moments: 1.0000E-06
- Language:** Español (selected), English
- Input file:** Buttons for file selection.
- Resultant calculation:** A large button to perform the calculation.

Pressing the button "Resultant calculation", the program will calculate the resultant for all load case. The results will be written in the output file and shown in the following form:

The screenshot shows the 'Resultante' application window with the title bar 'Resultante - C:\Temp\Example3.dat'. The window is divided into two tabs: 'Single load cases' (selected) and 'Combined load cases'. The 'Single load cases' tab displays a table with 7 columns: Case, Fx, Fy, Fz, Mx, My, and Mz. The table lists 22 single load cases, with the following data:

Case	Fx	Fy	Fz	Mx	My	Mz
4002	0.00	0.00	-87.46	0.000	53.407	0.000
4003	0.00	0.00	11.31	0.000	-7.223	0.000
4007	0.00	0.00	-26.47	0.000	-0.857	0.000
4009	0.00	0.00	-84.77	0.000	52.790	0.000
4013	0.00	0.00	-30.82	0.000	19.723	0.000
4014	0.00	0.00	-85.18	0.000	60.247	0.000
4015	0.00	0.00	48.41	0.000	-36.304	0.000
4019	0.00	0.00	68.33	0.000	-48.727	0.000
4020	0.00	0.00	-112.02	0.000	79.921	0.000

A 'Return' button is located at the bottom right of the table.

Two tabs are present to select the view of single or combined load cases. Pressing the button return, the input data form will be appearing again, with the previous data loaded.

7.- Examples

Four examples have been defined to show how to create the general input data file for the program RESULTANTE and to validate future changes of the program. In these examples only changes the cut section, being the origin point and the input files from NASTRAN the same.

The origin point to calculate forces and moments of the resultant is, in all cases:

$X_o = (12540, 0, 3000)$ with the coordinated expressed in “mm”

The input files from MSC/NASTRAN are:

Filename of GRIDS: example_grid.dat
 Filename of FORCESs & MOMENTS:..... example_force.dat
 Filename of LOADs: example_load.dat

The data contained in these files are in the following units:

Coordinates:..... mm
 Forces: N
 Moments N·mm

The resultant will be expressed in terms of:

Forces: kN ==> Factor on forces 1.0E-03
 Moments kN·m ==> Factor on moments 1.0E-06

7.1 Example 1

There is not cut section. Then, the whole structure is considered.

Taking into account the above and adding a title according with that, the general input data file as well as the output results are listed below:

General input data file:

```
1 Example 1 for the program RESULTANTE.
  No cut section: Whole structure processed.
  Input data units: [L]=mm [F]=N [M]=N.mm
  Output results units: [F]=kN [M]=kN.m
2 Origin for calculation of forces and moments: (Xo,Yo,Zo)
  12540.0  0.0  3000.0
3 Cut section: Option / 0:(-) 1:(Xmin) 2:(Ymin) 3:(Zmin) 4:(Xa,Ya,m)
5:(Xa,Za,m) 6:(Alfa)
  0
4 Nastran data files: GRID / FORCE-MOMENT / LOAD
  example_grid.dat
  example_force.dat
  example_load.dat
5 Results: Factor(Forces) Factor(Moments) / Output file
  1.000E-03  1.000E-06
  example1.txt
```

Output results file:

```

*****
*                                     *
*   <<< Calculation of the resultant >>>   *
*                                     *
*****

Filename of GRIDs:           example_grid.dat
Filename of FORCESS & MOMENTS: example_force.dat
Filename of LOADs:          example_load.dat

Title:
=====

Example 1 for the program RESULTANTE.
No cut section: Whole structure processed.
Input data units: [L]=mm [F]=N [M]=N.mm
Output results units: [F]=kN [M]=kN.m

Origin for calculation of forces and moments resultant:
=====

Xo = 12540.0
Yo =    0.0
Zo = 3000.0

Cut section:
=====

There is no cut: The whole structure will be processed.

File of GRIDs:
=====

Total number of GRIDs: Read ..... 54
                      Useful ..... 54
                      Erroneous ..... 0

File of FORCEs and MOMENTs:
=====

Total number of FORCEs: Read ..... 1009
                      Useful ..... 1009
                      Erroneous ..... 0

Total number of MOMENTs: Read ..... 1
                      Useful ..... 1
                      Erroneous ..... 0

File of LOADs:
=====

Total number of LOADs read ..... 22

```

Output results file: (cont.)

Single load cases:

=====

Cards FORCE and MOMENT

Case	Fx	Fy	Fz	Mx	My	Mz
-----	-----	-----	-----	-----	-----	-----
4002	-105.53	25.58	-410.65	-13.20	-85.96	-2.65
4003	-88.56	-2.13	90.40	6.57	14.81	0.97
4007	-73.75	8.33	-329.22	4.48	-142.24	-2.87
4009	-102.00	-16.21	-472.00	2.17	-103.81	2.69
4013	-107.44	21.85	-191.38	28.65	-51.39	-15.25
4014	-104.02	14.39	-448.60	5.40	-84.89	-6.20
4015	-122.21	9.80	70.56	-3.85	-37.49	-3.12
4019	-35.89	6.99	148.47	12.37	-44.65	-3.79
4020	-207.65	47.23	-442.54	15.72	-51.96	-10.31
4022	-130.94	16.14	80.76	14.35	-42.04	-13.69
4024	-14.34	32.39	-36.21	20.29	-9.88	-14.23
4025	-248.05	22.07	-239.62	6.41	11.75	-9.71
4028	6.56	-0.03	-57.58	1.24	-23.46	-1.66
4029	-72.72	7.96	-45.75	-0.89	0.61	1.12
4030	-279.91	-0.63	-302.72	-0.05	-1.33	0.12
4031	147.31	0.52	-126.54	0.54	-84.73	-0.10
4032	-180.44	-0.66	-239.98	-0.07	-13.93	0.15
4033	267.78	73.29	-66.79	75.89	-365.92	-15.43
4041	-96.78	-33.19	-201.92	0.51	-20.33	6.14
4042	-94.33	-3.55	-382.12	3.66	-79.84	-0.46
4043	-94.86	-50.70	42.29	-19.28	-20.94	15.88
4044	-107.42	3.86	-192.41	35.66	-51.52	-9.02

Fx, Fy, Fz have been multiplied by 1.0E-03

Mx, My, Mz have been multiplied by 1.0E-06

Combines load cases:

=====

Cards LOAD

Case	Denomination	Fx	Fy	Fz	Mx	My	Mz
-----	-----	-----	-----	-----	-----	-----	-----
2	B0310702 (WR2)	-75.38	18.27	-293.32	-9.43	-61.40	-1.89
3	A0110706 (WR6)	-63.26	-1.52	64.57	4.69	10.58	0.69
7	B0110713 (WR13)	-52.67	5.95	-235.16	3.20	-101.60	-2.05
9	B0310717 (WR17)	-72.86	-11.58	-337.14	1.55	-74.15	1.92
13	A0132605 (FTJ5)	-76.74	15.60	-136.70	20.46	-36.71	-10.89
14	A0232606 (FTJ6)	-74.30	10.28	-320.42	3.85	-60.64	-4.43
15	B0532608 (FTJ8)	-87.29	7.00	50.40	-2.75	-26.78	-2.23
19	B0144109 (RTJ9)	-25.64	4.99	106.05	8.84	-31.89	-2.71
20	B0144110 (RTJ1)	-148.32	33.74	-316.10	11.23	-37.12	-7.36
22	B0242912 (RTJ1)	-93.53	11.53	57.68	10.25	-30.03	-9.78
24	A0152111 (FIN1)	-10.24	23.14	-25.87	14.49	-7.05	-10.16
25	A1222705 (IT5)	-177.18	15.76	-171.15	4.58	8.39	-6.93
28	C01N6B (N6B)	4.37	-0.02	-38.39	0.82	-15.64	-1.11
29	C01N8D (N8D)	-48.48	5.31	-30.50	-0.59	0.41	0.75
30	C01M01 (M01)	-186.61	-0.42	-201.81	-0.03	-0.89	0.08
31	C01M02 (M02)	98.21	0.35	-84.36	0.36	-56.49	-0.07
32	C01M06B (M06B)	-120.29	-0.44	-159.99	-0.05	-9.29	0.10
33	C01028N7/AH1 (	178.52	48.86	-44.52	50.59	-243.94	-10.29
41	C0510909 (W230	-69.12	-23.71	-144.23	0.37	-14.52	4.39
42	B2110721 (WR21	-67.38	-2.53	-272.94	2.62	-57.03	-0.33
43	C0210722 (WR22	-67.76	-36.21	30.21	-13.77	-14.96	11.34
44	FUNCTIONAL RUD	-76.73	2.75	-137.44	25.47	-36.80	-6.44

Fx, Fy, Fz have been multiplied by 1.0E-03

Mx, My, Mz have been multiplied by 1.0E-06

7.2 Example 2

In this example, the cut section is defined by the plane $X = 12540$. Then, only loads applied on GRID point for which $X \geq 12540$ will be considered.

Taking into account the above and adding a title according with that, the general input data file as well as the output results are listed below:

General input data file:

```

1 Example 2 for the program RESULTANTE.
  Cut section: From the infinite up to X=12540.
  Input data units: [L]=mm [F]=N [M]=N.mm
  Output results units: [F]=kN [M]=kN.m
2 Origin for calculation of forces and moments: (Xo,Yo,Zo)
  12540.0  0.0  3000.0
3 Cut section: Option / 0:(-) 1:(Xmin) 2:(Ymin) 3:(Zmin) 4:(Xa,Ya,m)
5:(Xa,Za,m) 6:(Alfa)
  1
  12540.0
4 Nastran data files: GRID / FORCE-MOMENT / LOAD
  example_grid.dat
  example_force.dat
  example_load.dat
5 Results: Factor(Forces) Factor(Moments) / Output file
  1.000E-03  1.000E-06
  example2.txt
    
```

Output results file:

```
*****
*
*   <<< Calculation of the resultant >>>   *
*
*****
```

```
Filename of GRIDs:           example_grid.dat
Filename of FORCESS & MOMENTS: example_force.dat
Filename of LOADs:          example_load.dat
```

```
Title:
=====
```

```
Example 2 for the program RESULTANTE.
Cut section: From the infinite up to X=12540.
Input data units: [L]=mm [F]=N [M]=N.mm
Output results units: [F]=kN [M]=kN.m
```

```
Origin for calculation of forces and moments resultant:
=====
```

```
Xo = 12540.0
Yo =    0.0
Zo = 3000.0
```

```
Cut section:
=====
```

```
From the infinite up to Xmin = 12540.0
```

```
File of GRIDs:
=====
```

```
Total number of GRIDs: Read ..... 54
                        Useful ..... 24
                        Erroneous ..... 0
```

```
File of FORCEs and MOMENTs:
=====
```

```
In the following card, a force is applied on a nonexistent GRID point:
FORCE      4009  400704      1000.0  0.000  0.000  -4.784
```

```
In the following card, a force is applied on a nonexistent GRID point:
FORCE      4009  400703      1000.0  0.000  0.000  -4.784
```

```
. . . . .
. . . . .
. . . . .
```

```
Total number of FORCEs: Read ..... 1009
                        Useful ..... 352
                        Erroneous ..... 657
```

```
Total number of MOMENTs: Read ..... 1
                        Useful ..... 0
                        Erroneous ..... 1
```

```
File of LOADs:
=====
```

```
Total number of LOADs read ..... 22
```

Output results file: (cont.)

Single load cases:

=====

Cards FORCE and MOMENT

Case	Fx	Fy	Fz	Mx	My	Mz
-----	-----	-----	-----	-----	-----	-----
4002	0.00	0.00	-87.46	0.000	53.407	0.000
4003	0.00	0.00	11.31	0.000	-7.223	0.000
4007	0.00	0.00	-26.47	0.000	-0.857	0.000
4009	0.00	0.00	-84.77	0.000	52.790	0.000
4013	0.00	0.00	-30.82	0.000	19.723	0.000
4014	0.00	0.00	-85.18	0.000	60.247	0.000
4015	0.00	0.00	48.41	0.000	-36.304	0.000
4019	0.00	0.00	68.33	0.000	-48.727	0.000
4020	0.00	0.00	-112.02	0.000	79.921	0.000
4022	0.00	0.00	49.26	0.000	-39.305	0.000
4024	0.00	0.00	-0.91	0.000	0.650	0.000
4025	0.00	0.00	-50.51	0.000	42.220	0.000
4028	0.00	0.00	-6.81	0.000	2.900	0.000
4029	0.00	0.00	-1.78	0.000	2.362	0.000
4030	0.00	0.00	-58.13	0.000	38.899	0.000
4031	0.00	0.00	-25.23	0.000	19.988	0.000
4032	0.00	0.00	-41.33	0.000	27.388	0.000
4033	0.00	0.00	-8.97	0.000	5.957	0.000
4041	0.00	0.00	-35.87	0.000	31.034	0.000
4042	0.00	0.00	-61.03	0.000	47.772	0.000
4043	0.00	0.00	29.10	0.000	-25.854	0.000
4044	0.00	0.00	-32.34	0.000	20.382	0.000

Fx, Fy, Fz have been multiplied by 1.0E-03

Mx, My, Mz have been multiplied by 1.0E-06

Combines load cases:

=====

Cards LOAD

Case	Denomination	Fx	Fy	Fz	Mx	My	Mz
-----	-----	-----	-----	-----	-----	-----	-----
2	B0310702 (WR2)	0.00	0.00	-62.47	0.000	38.148	0.000
3	A0110706 (WR6)	0.00	0.00	8.08	0.000	-5.159	0.000
7	B0110713 (WR13)	0.00	0.00	-18.90	0.000	-0.612	0.000
9	B0310717 (WR17)	0.00	0.00	-60.55	0.000	37.707	0.000
13	A0132605 (FTJ5)	0.00	0.00	-22.01	0.000	14.088	0.000
14	A0232606 (FTJ6)	0.00	0.00	-60.84	0.000	43.034	0.000
15	B0532608 (FTJ8)	0.00	0.00	34.58	0.000	-25.932	0.000
19	B0144109 (RTJ9)	0.00	0.00	48.81	0.000	-34.805	0.000
20	B0144110 (RTJ1)	0.00	0.00	-80.02	0.000	57.086	0.000
22	B0242912 (RTJ1)	0.00	0.00	35.19	0.000	-28.075	0.000
24	A0152111 (FIN1)	0.00	0.00	-0.65	0.000	0.464	0.000
25	A1222705 (IT5)	0.00	0.00	-36.08	0.000	30.157	0.000
28	C01N6B (N6B)	0.00	0.00	-4.54	0.000	1.933	0.000
29	C01N8D (N8D)	0.00	0.00	-1.19	0.000	1.575	0.000
30	C01M01 (M01)	0.00	0.00	-38.75	0.000	25.933	0.000
31	C01M02 (M02)	0.00	0.00	-16.82	0.000	13.326	0.000
32	C01M06B (M06B)	0.00	0.00	-27.55	0.000	18.259	0.000
33	C01028N7/AH1 (	0.00	0.00	-5.98	0.000	3.972	0.000
41	C0510909 (W230	0.00	0.00	-25.62	0.000	22.167	0.000
42	B2110721 (WR21	0.00	0.00	-43.60	0.000	34.123	0.000
43	C0210722 (WR22	0.00	0.00	20.78	0.000	-18.467	0.000
44	FUNCTIONAL RUD	0.00	0.00	-23.10	0.000	14.559	0.000

Fx, Fy, Fz have been multiplied by 1.0E-03

Mx, My, Mz have been multiplied by 1.0E-06

7.3 Example 3

In this example, the cut section is defined by the plane $Z - 3000 = 4.70463 * (X - 12540)$. Then, only loads applied on GRID point for which $Z \geq 3000 + 4.70463 * (X - 12540)$ will be considered.

Taking into account the above and adding a title according with that, the general input data file as well as the output results are listed below:

General input data file:

```

1 Example 3 for the program RESULTANTE.
  Cut section: Defined by the plane:  $Z - 3000 = 4.70463 * (X - 12540)$ .
  Input data units: [L]=mm [F]=N [M]=N.mm
  Output results units: [F]=kN [M]=kN.m
2 Origin for calculation of forces and moments: (Xo,Yo,Zo)
  12540.0 0.0 3000.0
3 Cut section: Option / 0:(-) 1:(Xmin) 2:(Ymin) 3:(Zmin) 4:(Xa,Ya,m)
5:(Xa,Za,m) 6:(Alfa)
  5
  12540.0 3000.0 4.70463
4 Nastran data files: GRID / FORCE-MOMENT / LOAD
  example_grid.dat
  example_force.dat
  example_load.dat
5 Results: Factor(Forces) Factor(Moments) / Output file
  1.000E-03 1.000E-06
  example3.txt

```

Output results file:

```
*****
*
*   <<< Calculation of the resultant >>>
*
*****
```

```
Filename of GRIDs:           example_grid.dat
Filename of FORCESS & MOMENTS: example_force.dat
Filename of LOADS:          example_load.dat
```

```
Title:
=====
```

```
Example 3 for the program RESULTANTE.
Cut section: Defined by the plane:  $Z - 3000 = 4.70463 * (X - 12540)$ .
Input data units: [L]=mm [F]=N [M]=N.mm
Output results units: [F]=kN [M]=kN.m
```

```
Origin for calculation of forces and moments resultant:
=====
```

```
Xo = 12540.0
Yo =    0.0
Zo = 3000.0
```

```
Cut section:
=====
```

```
Defined by the plane:  $Z - Z_a = M * (X - X_a)$ 
```

```
      Xa =12540.0
      Za = 3000.0
      M  = 4.7046
```

```
File of GRIDs:
=====
```

```
Total number of GRIDs: Read ..... 54
                        Useful ..... 23
                        Erroneous ..... 0
```

```
File of FORCES and MOMENTS:
=====
```

```
In the following card, a force is applied on a nonexistent GRID point:
FORCE      4009 400704      1000.0  0.000  0.000 -4.784
```

```
In the following card, a force is applied on a nonexistent GRID point:
FORCE      4009 400703      1000.0  0.000  0.000 -4.784
```

```
. . . . .
. . . . .
. . . . .
```

```
Total number of FORCES: Read ..... 1009
                        Useful ..... 352
                        Erroneous ..... 657
```

```
Total number of MOMENTS: Read ..... 1
                        Useful ..... 0
                        Erroneous ..... 1
```

```
File of LOADS:
=====
```

```
Total number of LOADS read ..... 22
```

Output results file: (cont.)

Single load cases:

=====

Cards FORCE and MOMENT

Case	Fx	Fy	Fz	Mx	My	Mz
-----	-----	-----	-----	-----	-----	-----
4002	0.00	0.00	-87.46	0.000	53.407	0.000
4003	0.00	0.00	11.31	0.000	-7.223	0.000
4007	0.00	0.00	-26.47	0.000	-0.857	0.000
4009	0.00	0.00	-84.77	0.000	52.790	0.000
4013	0.00	0.00	-30.82	0.000	19.723	0.000
4014	0.00	0.00	-85.18	0.000	60.247	0.000
4015	0.00	0.00	48.41	0.000	-36.304	0.000
4019	0.00	0.00	68.33	0.000	-48.727	0.000
4020	0.00	0.00	-112.02	0.000	79.921	0.000
4022	0.00	0.00	49.26	0.000	-39.305	0.000
4024	0.00	0.00	-0.91	0.000	0.650	0.000
4025	0.00	0.00	-50.51	0.000	42.220	0.000
4028	0.00	0.00	-6.81	0.000	2.900	0.000
4029	0.00	0.00	-1.78	0.000	2.362	0.000
4030	0.00	0.00	-58.13	0.000	38.899	0.000
4031	0.00	0.00	-25.23	0.000	19.988	0.000
4032	0.00	0.00	-41.33	0.000	27.388	0.000
4033	0.00	0.00	-8.97	0.000	5.957	0.000
4041	0.00	0.00	-35.87	0.000	31.034	0.000
4042	0.00	0.00	-61.03	0.000	47.772	0.000
4043	0.00	0.00	29.10	0.000	-25.854	0.000
4044	0.00	0.00	-32.34	0.000	20.382	0.000

Fx, Fy, Fz have been multiplied by 1.0E-03

Mx, My, Mz have been multiplied by 1.0E-06

Combines load cases:

=====

Cards LOAD

Case	Denomination	Fx	Fy	Fz	Mx	My	Mz
-----	-----	-----	-----	-----	-----	-----	-----
2	B0310702 (WR2)	0.00	0.00	-62.47	0.000	38.148	0.000
3	A0110706 (WR6)	0.00	0.00	8.08	0.000	-5.159	0.000
7	B0110713 (WR13)	0.00	0.00	-18.90	0.000	-0.612	0.000
9	B0310717 (WR17)	0.00	0.00	-60.55	0.000	37.707	0.000
13	A0132605 (FTJ5)	0.00	0.00	-22.01	0.000	14.088	0.000
14	A0232606 (FTJ6)	0.00	0.00	-60.84	0.000	43.034	0.000
15	B0532608 (FTJ8)	0.00	0.00	34.58	0.000	-25.932	0.000
19	B0144109 (RTJ9)	0.00	0.00	48.81	0.000	-34.805	0.000
20	B0144110 (RTJ1)	0.00	0.00	-80.02	0.000	57.086	0.000
22	B0242912 (RTJ1)	0.00	0.00	35.19	0.000	-28.075	0.000
24	A0152111 (FIN1)	0.00	0.00	-0.65	0.000	0.464	0.000
25	A1222705 (IT5)	0.00	0.00	-36.08	0.000	30.157	0.000
28	C01N6B (N6B)	0.00	0.00	-4.54	0.000	1.933	0.000
29	C01N8D (N8D)	0.00	0.00	-1.19	0.000	1.575	0.000
30	C01M01 (M01)	0.00	0.00	-38.75	0.000	25.933	0.000
31	C01M02 (M02)	0.00	0.00	-16.82	0.000	13.326	0.000
32	C01M06B (M06B)	0.00	0.00	-27.55	0.000	18.259	0.000
33	C01028N7/AH1 (	0.00	0.00	-5.98	0.000	3.972	0.000
41	C0510909 (W230	0.00	0.00	-25.62	0.000	22.167	0.000
42	B2110721 (WR21)	0.00	0.00	-43.60	0.000	34.123	0.000
43	C0210722 (WR22)	0.00	0.00	20.78	0.000	-18.467	0.000
44	FUNCTIONAL RUD	0.00	0.00	-23.10	0.000	14.559	0.000

Fx, Fy, Fz have been multiplied by 1.0E-03

Mx, My, Mz have been multiplied by 1.0E-06

7.4 Example 4

There is not cut section. Then, the whole structure is considered, but the resultant will be projected a swept angle of 78 degrees.

Taking into account the above and adding a title according with that, the general input data file as well as the output results are listed below:

General input data file:

```

1 Example 4 for the program RESULTANTE.
  No cut section: Whole structure processed. Results projected with 78°.
  Input data units: [L]=mm [F]=N [M]=N.mm
  Output results units: [F]=kN [M]=kN.m
2 Origin for calculation of forces and moments: (Xo,Yo,Zo)
  12540.0  0.0  3000.0
3 Cut section: Option / 0:(-) 1:(Xmin) 2:(Ymin) 3:(Zmin) 4:(Xa,Ya,m)
5:(Xa,Za,m) 6:(Alfa)
  6
  78.0
4 Nastran data files: GRID / FORCE-MOMENT / LOAD
  example_grid.dat
  example_force.dat
  example_load.dat
5 Results: Factor(Forces) Factor(Moments) / Output file
  1.000E-03 1.000E-06
  example4.txt

```

Output results file:

```
*****
*                                     *
*   <<< Calculation of the resultant >>>   *
*                                     *
*****
```

```
Filename of GRIDs:           example_grid.dat
Filename of FORCESS & MOMENTS: example_force.dat
Filename of LOADS:           example_load.dat
```

```
Title:
=====
```

```
Example 4 for the program RESULTANTE.
No cut section: Whole structure processed. Results projected with 78°.
Input data units: [L]=mm [F]=N [M]=N.mm
Output results units: [F]=kN [M]=kN.m
```

```
Origin for calculation of forces and moments resultant:
=====
```

```
Xo = 12540.0
Yo =    0.0
Zo = 3000.0
```

```
Cut section:
=====
```

```
There is no cut: The whole structure will be processed.
The resultant will be projected with an angle of 78.0°
```

```
File of GRIDs:
=====
```

```
Total number of GRIDs: Read ..... 54
                        Useful ..... 54
                        Erroneous ..... 0
```

```
File of FORCES and MOMENTS:
=====
```

```
Total number of FORCES: Read ..... 1009
                        Useful ..... 1009
                        Erroneous ..... 0
```

```
Total number of MOMENTS: Read ..... 1
                        Useful ..... 1
                        Erroneous ..... 0
```

```
File of LOADS:
=====
```

```
Total number of LOADS read ..... 22
```


Output results file: (cont.)

Single load cases:

=====

Cards FORCE and MOMENT

Case	Fx	Fy	Fz	Mx	My	Mz
-----	-----	-----	-----	-----	-----	-----
4002	3.08	108.54	-410.65	-86.83	-4.96	-2.65
4003	-20.50	86.18	90.40	15.85	-3.34	0.97
4007	-7.19	73.86	-329.22	-138.20	-33.96	-2.87
4009	-37.06	96.40	-472.00	-101.09	-23.71	2.69
4013	-0.97	109.63	-191.38	-44.31	-38.71	-15.25
4014	-7.55	104.73	-448.60	-81.92	-22.93	-6.20
4015	-15.82	121.58	70.56	-37.47	-4.03	-3.12
4019	-0.63	36.56	148.47	-41.10	-21.38	-3.79
4020	3.03	212.93	-442.54	-47.56	-26.18	-10.31
4022	-11.44	131.44	80.76	-38.14	-22.78	-13.69
4024	28.70	20.76	-36.21	-5.44	-21.90	-14.23
4025	-29.99	247.22	-239.62	12.83	-3.83	-9.71
4028	1.33	-6.42	-57.58	-22.69	-6.09	-1.66
4029	-7.33	72.79	-45.75	0.42	0.99	1.12
4030	-58.82	273.66	-302.72	-1.31	-0.23	0.12
4031	31.14	-143.98	-126.54	-82.76	-18.15	-0.10
4032	-38.16	176.36	-239.98	-13.64	-2.83	0.15
4033	127.36	-246.69	-66.79	-342.14	-150.31	-15.43
4041	-52.59	87.76	-201.92	-19.78	-4.73	6.14
4042	-23.08	91.53	-382.12	-77.33	-20.18	-0.46
4043	-69.31	82.25	42.29	-24.49	14.50	15.88
4044	-18.56	105.88	-192.41	-42.98	-45.59	-9.02

Fx, Fy, Fz have been multiplied by 1.0E-03

Mx, My, Mz have been multiplied by 1.0E-06

Combines load cases:

=====

Cards LOAD

Case	Denomination	Fx	Fy	Fz	Mx	My	Mz
-----	-----	-----	-----	-----	-----	-----	-----
2	B0310702 (WR2)	2.20	77.53	-293.32	-62.02	-3.54	-1.89
3	A0110706 (WR6)	-14.64	61.56	64.57	11.32	-2.39	0.69
7	B0110713 (WR13)	-5.13	52.76	-235.16	-98.71	-24.26	-2.05
9	B0310717 (WR17)	-26.47	68.86	-337.14	-72.21	-16.93	1.92
13	A0132605 (FTJ5)	-0.69	78.31	-136.70	-31.65	-27.65	-10.89
14	A0232606 (FTJ6)	-5.40	74.81	-320.42	-58.51	-16.38	-4.43
15	B0532608 (FTJ8)	-11.30	86.84	50.40	-26.76	-2.88	-2.23
19	B0144109 (RTJ9)	-0.45	26.11	106.05	-29.36	-15.27	-2.71
20	B0144110 (RTJ1)	2.16	152.09	-316.10	-33.97	-18.70	-7.36
22	B0242912 (RTJ1)	-8.17	93.88	57.68	-27.24	-16.27	-9.78
24	A0152111 (FIN1)	20.50	14.83	-25.87	-3.89	-15.64	-10.16
25	A1222705 (IT5)	-21.42	176.59	-171.15	9.16	-2.74	-6.93
28	C01N6B (N6B)	0.89	-4.28	-38.39	-15.13	-4.06	-1.11
29	C01N8D (N8D)	-4.89	48.52	-30.50	0.28	0.66	0.75
30	C01M01 (M01)	-39.21	182.44	-201.81	-0.87	-0.15	0.08
31	C01M02 (M02)	20.76	-95.99	-84.36	-55.18	-12.10	-0.07
32	C01M06B (M06B)	-25.44	117.57	-159.99	-9.10	-1.89	0.10
33	C01028N7/AH1 (	84.91	-164.46	-44.52	-228.09	-100.21	-10.29
41	C0510909 (W230	-37.56	62.69	-144.23	-14.13	-3.38	4.39
42	B2110721 (WR21	-16.49	65.38	-272.94	-55.24	-14.41	-0.33
43	C0210722 (WR22	-49.51	58.75	30.21	-17.50	10.36	11.34
44	FUNCTIONAL RUD	-13.26	75.63	-137.44	-30.70	-32.57	-6.44

Fx, Fy, Fz have been multiplied by 1.0E-03

Mx, My, Mz have been multiplied by 1.0E-06